

Rocket Nozzle Simulation

OpenFOAM v2012 · rhoCentralFoam · Axisymmetric wedge · Compressible supersonic flow

A parametric study of flow through a De Laval (converging-diverging) rocket nozzle, comparing an idealised hot-gas model against a reacting LOX/kerosene combustion case. Both cases use the same axisymmetric wedge geometry and are run with density-based compressible solvers capable of capturing the transonic throat, supersonic expansion, and shock structure at the nozzle exit.

Cases

Case	Solver	Inlet model	Status
Hot gas (frozen composition)	rhoCentralFoam	Combustion products approximated as a perfect gas (M=22 g/mol, gamma=1.2)	Complete
LOX/kerosene reacting flow	TBD	Full combustion chemistry with species transport	In progress

Nozzle Geometry

All cases share the same axisymmetric conical De Laval profile:

Parameter	Value
Inlet radius	0.150 m
Throat radius	0.050 m
Exit radius	0.1414 m
Throat position	x = 0.15 m
Total nozzle length	0.50 m
Converging area ratio	9:1
Diverging area ratio	8:1
Design exit Mach (gamma=1.2)	~3.2

The mesh is a 5-degree wedge (axisymmetric) built with blockMesh, 40 radial cells x 150 axial cells = 6,000 cells total. The two-block structure (converging + diverging) uses wedge boundary patches on the front and back faces and an empty patch on the degenerate axis line.

Hot Gas Case

Setup

Combustion products of LOX/kerosene are approximated as a single-species perfect gas with properties representative of the chamber exit composition:

Property	Value	Notes
Molecular weight	22 g/mol	Mix of CO ₂ , H ₂ O, CO, H ₂
gamma	1.2	Tri-atomic dominated at high T
Cp	2267 J/(kg K)	Derived from gamma, M
Dynamic viscosity	1e-4 Pa s	Approximate at 3000 K
Chamber total pressure	3 MPa	Approximate LOX/kerosene adiabatic flame temperature
Chamber total temperature	3000 K	
Ambient back pressure	101,325 Pa	Sea level — nozzle is over-expanded at design

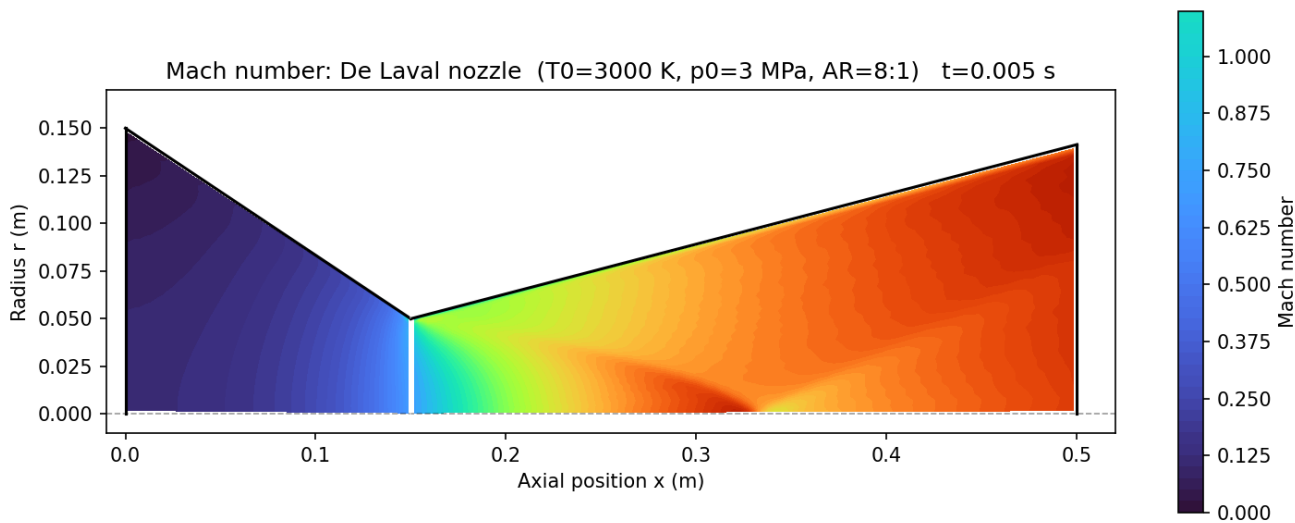
Boundary conditions: totalPressure and totalTemperature at the inlet, waveTransmissive at the outlet. The case is run laminar (viscous effects are secondary for this geometry and Reynolds number).

Solver: rhoCentralFoam with Kurganov-Tadmor flux scheme, MUSCL reconstruction (vanLeer for rho and T, vanLeerV for U), CFL = 0.1. Initialised with an isentropic pressure and temperature ramp along x to avoid the

numerical temperature inversion that occurs when starting from uniform chamber conditions against a near-vacuum outlet.

Results

Mach number contour



The flow accelerates from rest at the inlet, reaches sonic conditions (Mach 1) at the throat ($x = 0.15\text{ m}$), then expands supersonically through the diverging section to a peak exit Mach of **3.22**, matching the isentropic design prediction of 3.2 to within 1%.

Quantity	Simulated	Isentropic theory
Exit Mach	3.22	3.20
Exit static T	1488 K	1482 K
Peak exit velocity	2667 m/s	2637 m/s

The oblique feature visible near $x = 0.2\text{ m}$ is an expansion fan originating from the sharp throat corner — a real compressible flow effect, not a numerical artefact. The nozzle is over-expanded at sea level (design exit pressure $\sim 43.5\text{ kPa}$ vs 101.3 kPa ambient), which would produce oblique shocks outside the nozzle exit in a full domain; those are outside the current computational domain.

LOX/Kerosene Case

(In progress) — Full reacting flow with species transport and combustion chemistry, replacing the frozen-composition inlet condition with actual LOX/kerosene combustion. Same nozzle geometry.

Workflow

```
blockMesh (5-degree wedge, 6000 cells) | rhoCentralFoam (density-based, explicit, CFL=0.1) |
Post-processing: Mach = |U| / sqrt(gamma * R_spec * T)
```

Key solver notes:

- rhoCentralFoam requires fluxScheme Kurganov and vanLeer interpolation schemes — these are not inherited from fvSchemes defaults and must be set explicitly.
- totalTemperature BC in v2012 requires a gamma entry in the field file, not just in thermophysicalProperties.
- Initialising with uniform chamber conditions causes temperature inversion in the first timestep due to the large pressure gradient across the domain. An isentropic ramp initialisation resolves this.
- The wedge block vertex ordering must ensure direction 3 (the extrusion direction, v0 to v4) is a non-degenerate edge — starting both groups with a wall vertex rather than an axis vertex avoids the "inward-pointing faces" error from the degenerate zero-length axis edge.

Limitations

- Conical nozzle profile — a bell or Rao-optimised profile would reduce exit flow divergence angle and improve specific impulse.
- Adiabatic wall assumption — no heat transfer to the nozzle wall.
- Laminar — boundary layer transition and turbulent mixing near the wall are not modelled.
- 2D axisymmetric — no three-dimensional instabilities or asymmetric shock structures captured.
- The over-expanded exit shock structure forms outside the domain.

Software Stack

Tool	Version	Purpose
OpenFOAM	v2012	CFD solver
ParaView	5.11	Post-processing
Python / matplotlib	3.x	Mach contour plot
Ubuntu (WSL2)	24.04	OS

Repository

Part of the [OpenFOAM Portfolio](#), a collection of CFD simulations demonstrating thermal and fluid simulation skills relevant to thermal engineering roles.